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9. (a) The elements of the p-block show a variety of oxidation states within the same group, whilst some elements of the d-block can show an even wider range of oxidation states.

- Some oxides of group 4 have an oxidation state of +4 such as CO_2, SiO_2 and PbO_2 and others have an oxidation state of +2 such as CO and PbO. There is no stable oxide with the formula SiO.
- Some chlorides of group 5 have a +5 oxidation state such as PCl_5 and some have an oxidation state of +3 such as NCl_3 and PCl_3. There is no stable chloride with the formula NCl_5.
- Manganese forms a range of oxides including MnO, Mn_2O_3 and MnO_2 which have oxidation states of +2, +3 and +4 respectively.

Explain why these elements can form the oxidation states shown and why SiO and NCl_5 do not form. [6 QER]

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(b) Tin(II) ions, Sn^{2+} (aq), can be used as reducing agents for a range of metal ions. In one study of the reduction of Pt^{4+} ions to Pt^{2+} ions in an acid solution, the reaction was found to be first order with respect to both Pt^{4+} and Sn^{2+} with a rate constant of $895\,mol^{-1}\,dm^{3}\,s^{-1}$ at 297 K.

(i) Write a rate equation for this reaction. [1]

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(ii) A student suggests studying this reaction using sampling and quenching with samples taken every 30 seconds for 5 minutes. Explain why this would not be an appropriate sampling interval. [2]

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(c) (i) A second student decides to use colorimetry to study the rate of this reaction. Explain what the student should consider when choosing an appropriate wavelength of light for their colorimetry experiment. [1]

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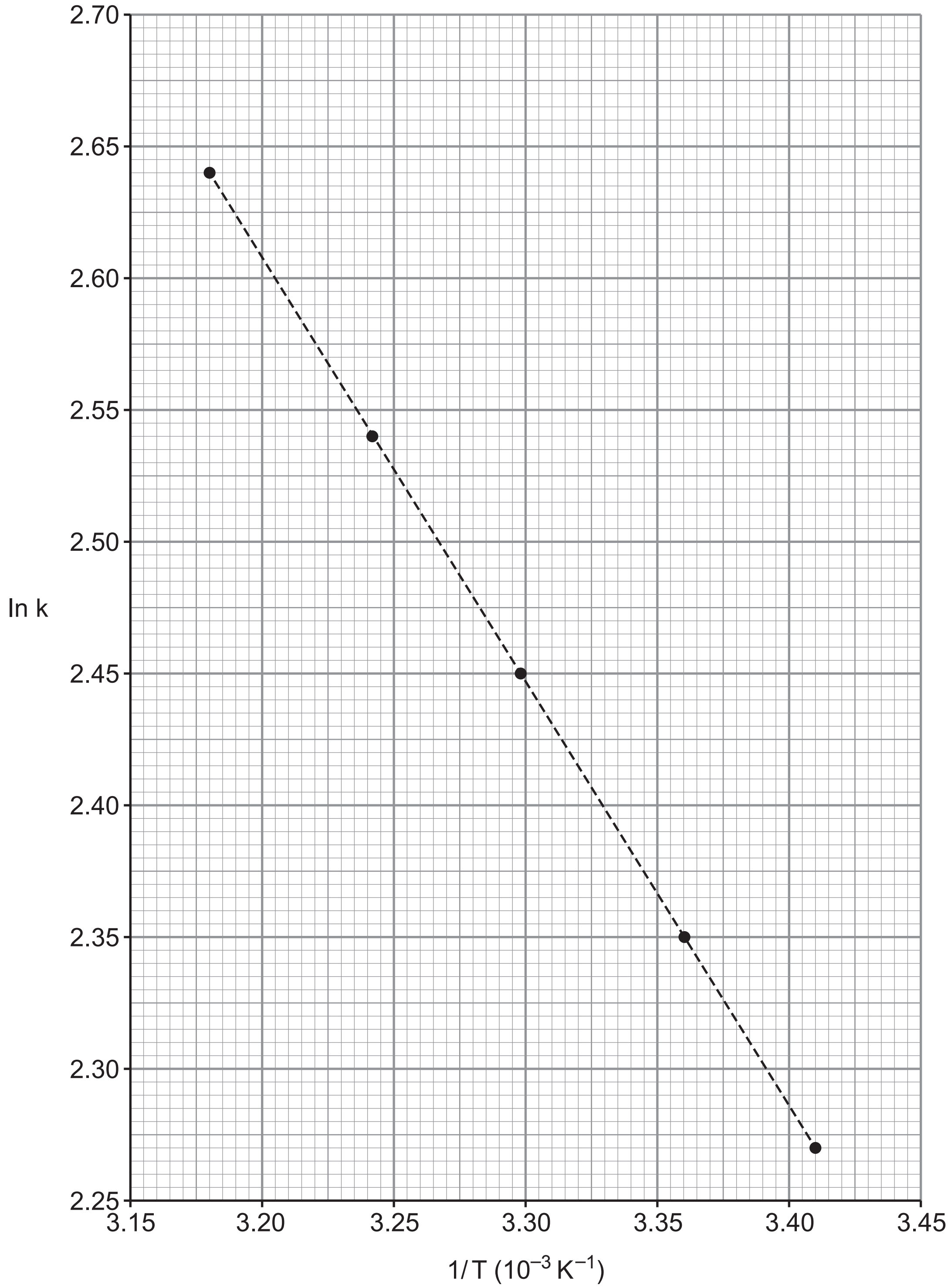
(ii) The value of the rate constant was measured at five different temperatures to attempt to find the activation energy.

$$\ln k = \ln A - (E_a/RT)$$

A graph was plotted of (ln k) against (1/Temperature) and this is shown below.



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Find the activation energy of the reaction. [3]

activation energy, E_a = ... $kJ\,mol^{-1}$

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Turn over.